Channels

# One citizen, many channels: app, web, sorting guide, collector mode, smart bins

## Citizen

- Built — Mobile app (Expo)
- Built — Citizen website
- Built — Photo and voice note (typing fallback)
- Built — Picture-based sorting guide

## One backend, one database

- FastAPI · one API contract
- PostgreSQL / Supabase (SQLite locally)
- Device-key API for bins and scales
- AI suggestions, human confirms

## City and field

- Built — Collector mode, same app
- Built — Command Center for staff
- Simulated — Smart bin over GSM
- Simulated — Hub scale at a collection point

**The same report and the same record, whichever door you use.**

Built · Simulated · Designed · Planned — Status today, from the repo

CityCycle

# CityCycle: households, price list, hubs and an impact ledger

## Household

Picture sorting guide, pickup booking, free or paid, estimate before pickup

Built

## District price list

The district sets prices. A material without a price gets no estimate. All prices shown are demo prices.

Built Demo values

## Collection hubs

A scale records verified weight. The API route exists; the prototype keeps weighings in a local queue.

Built Simulated

## Impact ledger

| | |
|---|---|
| 13.5 kg | diverted |
| 51.5 EGP | value of weighed material |
| 48 EGP | recorded as paid to households |
| 5 / 1 | pickups / job |
| 0.0135 t | tonnes |

Demo values Verified in tests

$CO_2$ is withheld until each material has a sourced factor. Amounts are recorded; no money moves.

# Collector mode: clear jobs, verified weight, recorded earnings

## What the collector gets

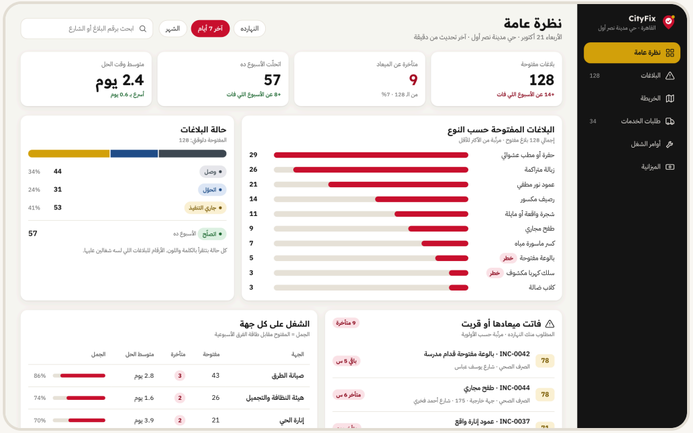
| | |
|---|---|
| Nearby jobs on a map, approximate pin only | Verified in tests |
| Verified weights, weighings are idempotent | Verified in tests |
| Earnings record and payout list | Built |
| Works on weak networks: actions queue and sync later | Built |
| Picture-based, low-literacy interface | Designed |

## How trust is built

| | |
|---|---|
| First accept wins; the second gets a 409 | Verified in tests |
| The household confirms before it closes | Verified in tests |
| Collector applies; the district reviews | Built |
| Staff see the district, not the home address | Verified in tests |
| Training, insurance and ID checks | Planned |

Collector mode is built in the mobile app (apps/mobile/src/app/collector), mock mode by default. Not yet field-tested with real collectors.

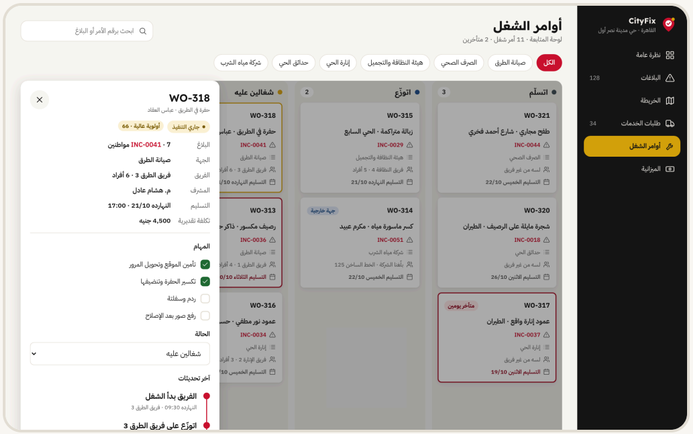
# The Command Center: the district sees, ranks and decides



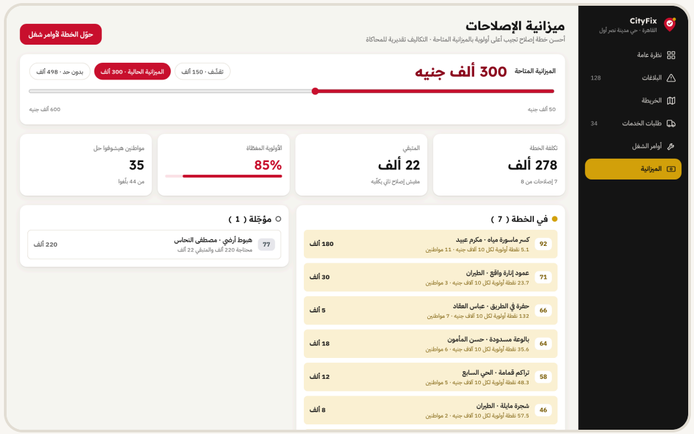

Overview




Incident detail




Work orders




Budget planner

## Command Center

| | |
|---|---|
| Incidents ranked; the priority score is explained | Built |
| Work orders and department assignment | Built |
| Budget planner picks the best repairs in budget | Built |
| Map: incidents, bins, dumping hotspots | Built |
| Bin fill forecast and route preview | Built |

Screens from the design files. Forecast, route and bin screens exist in apps/admin-web (vitest-tested); not yet run on a deployed API.

CityBrain

# CityBrain: AI that suggests, and a human who decides

1. Report with text, photo or voice
2. Gemini suggests category and severity
3. No Gemini: keyword fallback
4. A human confirms the suggestion
5. An explained priority score

## Safeguards

| | |
|---|---|
| AI only suggests; a human confirms | Built |
| Deterministic keyword fallback when Gemini is unavailable | Built |
| Photo analysis is off by default until the privacy gate is passed | Built |
| No accuracy claim: 33 synthetic cases, top-1 90.9%, top-3 100%, not a benchmark | Verified in tests |

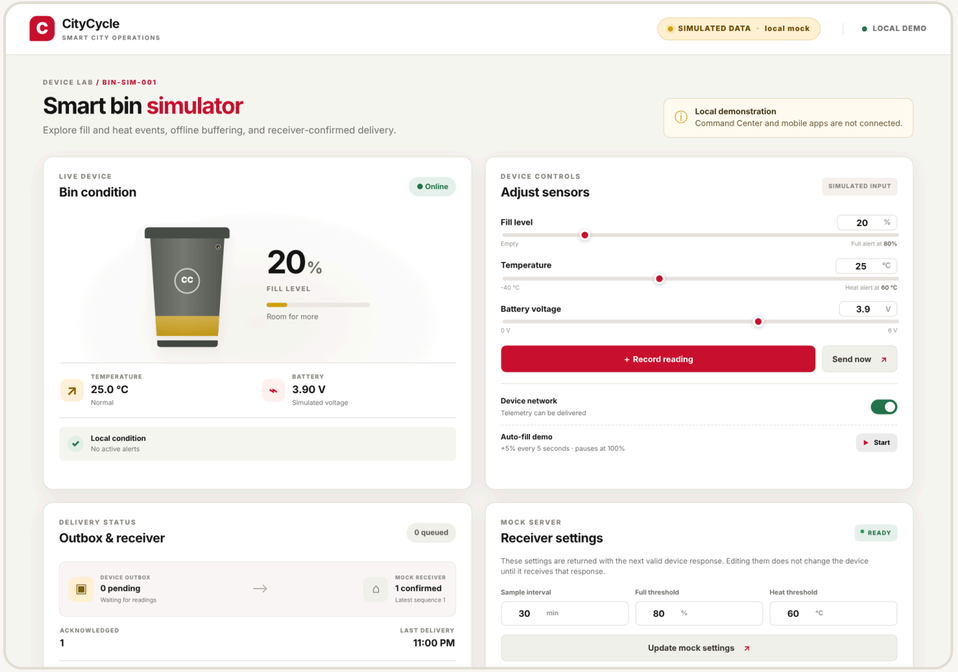
## More in CityBrain

| | |
|---|---|
| Linear fill forecast: 3+ readings over 1+ hour | Verified in tests |
| OR-Tools capacity-constrained route preview (pickups only; bins not stops yet) | Verified in tests |
| Dumping hotspots from repeat reports | Built |
| Gemini quality on Egyptian photos and voice | To verify |

# Smart bin and hub scale: designed and simulated, honestly labelled

## The smart bin and hub scale

| | |
|---|---|
| Ultrasonic fill-level sensor and a temperature probe | Designed |
| ESP32 deep-sleeps and wakes on a timer | Simulated |
| 18650 cell with solar charging | Designed |
| SIM800L modem (2G only), LTE as the alternative | Designed |
| Device key plus sequence number; retries never duplicate | Verified in tests |
| Network drops: readings buffer and retry later | Simulated |




PC simulator (hardware/simulator), local mock: real screenshot, not connected to the API

**Simulated: no physical unit has been built or field-tested yet**

Sources: hardware/README.md, hardware/schematics/smart-bin.svg, docs/field-brief/field-deployment-brief.md, backend/tests/test_bins.py.

Architecture

# One backend, three clients, devices with a per-device key, AI that only suggests

| | | |
|---|---|---|
| Mobile app: citizen and collector — Built | Citizen website — Built | Command Center — Built |

▼ ▲

**One API contract · same endpoints and shapes for every client**

| | | |
|---|---|---|
| PostgreSQL + PostGIS on Supabase; SQLite locally; Alembic migrations; row-level security — Built | Gemini triage with keyword fallback, linear forecast, OR-Tools route preview — Built | ESP32 smart bin and hub scale post to /iot with a device key — Simulated |

| | | | |
|---|---|---|---|
| Citizen OTP login; staff accounts | Per-device key; (binId, seq) is unique | Idempotency keys on writes | Redacted staff view; approximate pin |

Source: docs/architecture.md and docs/diagrams/system-architecture.svg. Production deployment is not yet verified.

# Technical rigor: what tests verify, and what is still open

**430**
backend tests passed (9 skipped)

**66**
simulator tests OK

**33**
synthetic AI smoke cases, not a benchmark

**1 / folder**
CI workflow per app folder

## Verified in tests

| | |
|---|---|
| First accept wins; weighings are idempotent | Verified in tests |
| Duplicate or conflicting readings are refused | Verified in tests |
| Staff see a district; collectors cannot locate a home | Verified in tests |
| Fill forecast and route partitioning | Verified in tests |

## Still to verify

| | |
|---|---|
| Sensor accuracy in heat and dust, GSM and battery | To verify |
| Forecast accuracy on real bins | To verify |
| Gemini quality on consented Egyptian data | To verify |
| A live deployment, and OR-Tools in production | To verify |

Counts recorded 2026-10-23 (docs/evidence.md); re-run before presenting. Reproduce: backend pytest, python -m unittest in hardware/simulator.

# Impact: measured from platform data, not estimated

## What we measure, and where the number comes from

| | | |
|---|---|---|
| **kg diverted from landfill** | Confirmed weighings (collector or hub scale) | Built |
| **EGP earned by collectors** | Weighings at the district price list, and payouts | Built |
| **Pickup response time** | Request to acceptance, and to completed weighing | To verify |
| **Bins that never overflow** | Share of time under the full threshold (threshold agreed with the district) | Simulated |
| **Dumping hotspots cleared** | Hotspots detected vs cleared, with time to clear | To verify |
| **$CO_2$ avoided (secondary)** | Withheld until each material has a sourced emission factor | To verify |

## Demo seed values today

| | |
|---|---|
| **13.5 kg** | diverted (seed) |
| **51.5 EGP** | weighed value (seed) |
| **48 EGP** | recorded as paid (seed) |
| **5** | pickups (seed) |
| **0.0135 t** | tonnes (seed) |

Demo values

No target, savings or income figure is promised before real pilot data.

# Feasibility: local parts, and the field risks we already know

## Parts in EGP, dated 2026-10-23 (partial)

| | |
|---|---|
| ESP32 | 280 EGP |
| DS18B20 probe | 65 EGP |
| Resistor kit | 95 EGP |
| TP4056 charger | 15 EGP |
| 18650 cell | 165 EGP |
| Cell holder | 8 EGP |
| Schottky diode | 1.25 EGP |
| 5 V boost | 25 EGP |
| P-MOSFET ×2 | 7 EGP |
| SIM800L GSM | 300 EGP |
| Cable glands ×3 | 12 EGP |

**11 sourced lines = 973.25 EGP, one-time. Not a per-bin cost: probe, solar panel, enclosure and 14 more lines still need quotes.**

## Field readiness

| | |
|---|---|
| Heat, dust and street washing: sealed box, breather vent, shaded sensor | Designed |
| Weak or no GSM: 32-reading buffer, external antenna, LTE option | Designed |
| Theft: security screws, nothing visible, tamper switch | Designed |
| Monthly cleaning; swap a pre-tested tray | Designed |
| Bench and field tests on a real unit | To verify |

Sources: hardware/bom/bom.csv, docs/field-brief/field-deployment-brief.md. Design only: no unit has been built or placed on a street.

Proposed development and pilot plan

# Closing the Circular Economy Loop: from waste collection to verified recovery

**1** **Household sorts waste** — Built

**2** **Collector picks it up** — Built

**3** **Hub verifies weight** — Simulated

**4** **Recycler receives materials** — Planned

**5** **Payment and recovery are recorded** — Planned

## What we would record

- The receiving recycler, accepted weight, material quality and sale value.
- “Collected” and “confirmed recovered” tracked separately.
- A clear record of each handover, visible to households and collectors.

## Two states, never mixed

**Collected**
Picked up and weighed at the hub

**Confirmed recovered**
Accepted by a recycler, with weight and quality

Only the second state counts as recovery. Today the platform records the first.

Proposed development: recycler receipt and confirmed recovery are not built yet. Weighing, pickups and amounts recorded are built (see slides 9 and 10).

Proposed development and pilot plan

# Empowering Egypt’s Informal Collectors

## Build on the Zabbaleen’s expertise

| | |
|---|---|
| Partner with existing collectors, collection hubs and cooperatives. | Built |
| Provide nearby jobs, transparent prices and weighing records. | Built |
| Distinguish manual weights from calibrated scale measurements. | Planned |
| Show net earnings: sales revenue minus household payments and operating costs. | Planned |
| Provide simple Arabic, icons, voice guidance and offline access. | Designed |
| Include a way to question a weight or payment. | Planned |

## Net earnings, in plain terms

**Sales revenue**

**– household payments**

**– operating costs**

---

**= net earnings**

Technology should strengthen collectors’ existing work, protect their role and help them earn more.

Proposed development. Status tags show what exists today; Arabic UI, jobs and recorded weights are in the app, the rest is planned for the pilot.

Proposed development and pilot plan

# A Practical Pilot with Measurable Results

## Start locally, measure, then expand

- Begin with one neighbourhood, participating collectors and a recycling partner.
- Use existing bins and hubs; add sensors where they provide clear value.
- Demonstrate the full collection-to-recycler journey.
- Validate hardware accuracy, connectivity, maintenance and total cost before expansion.

## Measure

Every pilot figure is measured on site. No savings or income number is promised before real data.

| Economic benefit | Environmental benefit |
|---|---|
| Collector net earnings per working hour | Confirmed material recovery |
| Cost per kilogram recovered | Material contamination |
| Time spent travelling and collecting | Overflow duration and dumping-clearance time |

**“CityCycle connects households, informal collectors and recyclers through transparent prices, traceable weights and reliable collection—supported by AI and smart-bin technology.”**

Evaluation framework

# How each of the five criteria is answered, and where to see it

| | | | |
|---|---|---|---|
| 25 | Local context and impact | Arabic-first, picture sorting, pay per kg for collectors, bins and hotspots for the district | Slides 2, 4–7, 16 |
| 25 | Technical rigor and MVP | Working app, website and Command Center; tests; device-key telemetry; simulator | Slides 8–15 |
| 20 | Innovation and interdisciplinarity | Software, AI, GIS, IoT hardware and a hub scale in one loop | Slides 5, 12–13 |
| 20 | Economic and field feasibility | Local parts with dated EGP quotes, field risks and a pilot plan | Slides 17, 19 |
| 10 | Technical communication and pitch | Architecture, docs, status tags and a 3-minute story | All slides |

Points are the SolveThe17 2.0 evaluation framework (total 100). The slides show evidence; the scoring is the judges' call.

Next steps

# Team, roadmap and a link to try it

## Team

Moataz Hendy
Mohamed Wageeh
Hesham El-Kateb
Ziad Rabie
Abdulrahman Gomaa

## Try it now

Report it. We follow it.

## Roadmap

### Today

Working app, website, Command Center; bins simulated

### Pilot

One real bin and a district price list

### Verify

Dated quotes, agreed thresholds, field tests

### Deploy

Verified deployment, expand after open items close

Scan to open the citizen site and report something

**Citizen web** · city-fix-trojans.vercel.app

**Command Center** · city-fix-admin.vercel.app

**API** · city-fix-api-site.vercel.app

Links will be updated before the pitch. A live deployment is not yet verified in the repo.

# Appendix A1: the API contract, grouped by who calls it

### Citizen

`POST /otp` `POST /verify` `GET /me` `GET /me/impact` `POST /reports` `GET /incidents` `POST /incidents/{id}/follow` `POST /service-requests` `GET /map/incidents` `GET /notifications`

### Recycling

`GET /recycling/materials` `GET /material-prices` `POST /recycling/pickups` `POST /recycling/pickups/{id}/confirm` `POST /recycling/pickups/{id}/cancel` `GET /recycling/summary`

### Collector

`POST /collectors/apply` `GET /collector/jobs` `POST /collector/jobs/{id}/accept` `POST /collector/jobs/{id}/weighings` `POST /collector/jobs/{id}/complete` `GET /collector/earnings`

### Staff and admin

`GET /dashboard` `PATCH /incidents/{id}` `POST /incidents/{id}/triage/confirm` `POST /work-orders` `POST /budget/optimize` `GET /admin/bins/forecast` `GET /admin/hotspots` `POST /admin/recycling/routes/optimize` `PUT /material-prices`

### Devices (device key)

`POST /iot/bins/{id}/readings` `POST /iot/hubs/{id}/weighings` `POST /bins/{id}/report-full` `GET /bins`

Selected endpoints read from backend/app/routers. Full contract: docs/api-contract.md.

# Appendix A2: the data model in four groups

### People

Profile · StaffAccount · Collector · Department · Notification · PushToken

### Reports and operations

Incident · Report · IncidentFollower · ServiceRequest · WorkOrder · ActivityLog · Category · BudgetPlan · AISuggestion

### CityCycle

RecyclingPickup · Weighing · MaterialPriceRow · SmartBin · BinReading · BinReport · HubWeighing

### Safety

IdempotencyKey · OTPChallenge · OTPThrottle

Tables from backend/app/models.py. PostgreSQL with PostGIS on Supabase; Alembic migrations; SQLite for local runs.

Who benefits

# Three groups each get something they can see and measure

## Citizens

Sort at home with a picture guide, book a pickup, see what material is worth, give it free or get paid, and watch reports get fixed

## Informal collectors (Zabbaleen)

Nearby jobs, verified weights, a record of earnings, pay per kg from a published district price list

## Districts

Bins emptied when full, shorter routes, visible dumping hotspots, tonnes diverted measured not estimated

From the repo vision. Benefits are what the platform records and shows; no savings or income figure is claimed before pilot data.

The demo story

# Request, accept, weigh, confirm, impact, and a bin that calls for collection

1

## Household requests a pickup

Verified in tests

`test_create_pickup_estimates_from_price_list`

2

## Collector accepts the job

Verified in tests

`test_first_accept_wins_and_weighing_rules`

3

## Collector weighs

Verified in tests

`test_full_flow`

4

## Household confirms

Verified in tests

`test_full_flow`

5

## Impact is recorded

Verified in tests

`test_impact_and_earnings`

**And the smart bin calls for collection**

A full or hot bin shows in the public bin list, and a citizen can report a full bin. Readings come from the simulator today.

Verified in tests

Simulated

Test names are in backend/tests/test_citycycle.py. Amounts are recorded only: no payment provider is integrated.

# Report, follow, and book a service from one app

## Citizen app

| | |
|---|---|
| Pick the type; emergency numbers up front | Built |
| Photo, voice note, location | Built |
| AI suggests, the citizen confirms | Built |
| Follow it; "me too" on the map | Built |




Home




Report




Follow it




CityServe

Screens: rendered from the team design files (mobile-Main, mobile-Report, mobile-Track, mobile-Services). The app runs on mock data by default.

# Appendix A3: privacy and consent as built today

| | |
|---|---|
| Staff see the district, not the household address | Verified in tests |
| Collectors see an approximate pin; exact home cannot be located from the jobs list | Verified in tests |
| Photo analysis off by default until tested on consented Egyptian samples | To verify |
| The smart bin has no camera and collects no personal data | Designed |
| Each device has a key that can be rotated or revoked | Verified in tests |
| OTP login; demo OTP mode is for demos only | Built |

Not a legal compliance claim. A full consent and data-retention policy is still to write.

# Appendix A4: risk register

| | | |
|---|---|---|
| **Sensor accuracy in heat and dust** | Bench and field tests on a real unit | To verify |
| **GSM or 2G not available** | LTE Cat-1 option; test signal at the real location | To verify |
| **Theft and vandalism** | Hidden screws, nothing visible, tamper switch | Designed |
| **AI suggests the wrong category** | A human confirms; keyword fallback | Built |
| **Demo prices and thresholds** | The district sets real prices and thresholds | Planned |
| **No payment integration** | Amounts recorded only; integrate a provider later | Planned |
| **Collectors may not adopt it** | Pilot with collectors; picture UI; verified pay | To verify |
| **A live deployment is unverified** | Runbook in docs/evidence.md before the pitch | To verify |

Mitigation tag = status of the mitigation today. Sources: docs/judges-qa.md, docs/evidence.md, docs/field-brief.

SolveThe17 2.0

Track 4 · SDG 11

CityFix

# A cleaner city, faster services, and waste that becomes value.

## Unlocking Egypt's Waste Economy

**Report it. We follow it.**



Moataz Hendy · Mohamed Wageeh · Hesham El-Kateb
Ziad Rabie · Abdulrahman Gomaa

# Lots of waste, value lost, and workers who go unseen

**~20M**

tonnes of municipal solid waste a year

Theme fact EEAA

**80%**

up to, recovered by the Zabbaleen

Theme fact SolveThe17 brief

## The gap

No structured framework connects households, collectors and the city, so the recovery stays informal and unmeasured.

Figures are quoted from the SolveThe17 2.0 Track 4 theme brief; we have not measured them ourselves.

How might we establish a structured circular economy framework in Egypt that transforms waste streams into economic value while elevating informal sector workers and improving urban environmental health?

## Our answer: one platform that connects households, collectors and the city, in three parts.

### 1 A structured framework

| | |
|---|---|
| Published price list and pickup requests | Built |
| Collector mode: nearby jobs and weighing | Built |
| Smart bins and hub scale | Simulated |

### 2 Economic value

| | |
|---|---|
| Pay per kg from a price list (demo prices) | Built |
| Impact ledger: kg and $CO_2$ avoided | Built |
| Measured in a field pilot | Planned |

### 3 Workers and urban health

| | |
|---|---|
| Collectors see jobs and earnings openly | Built |
| Street reports followed until fixed | Built |
| Collector verification and training | Planned |

Built · Simulated · Designed · Planned — Status today, from the repo

The solution

# One platform for city life: six pillars, not only recycling

## FixFirst

Report problems with photo, voice and GPS; follow each one until it is fixed

Built

## CityServe

Book city services from the same app

Built

## CityCycle

Recycling pickups, collector mode, material prices, impact, smart bins

Built Simulated

## CityOps

Work orders and department assignment for the district

Built

## CityPulse

Live map: incidents, smart bins and dumping hotspots

Built

## CityBrain

AI triage, fill forecast and route preview. AI only suggests

Built To verify

Built Simulated Designed Planned Status today, from the repo